

Analysis:

- ↳ Examine the problem
- ↳ Purpose

Purpose → Scope → Boundaries → Functional Requirements

Input → Processes → Output.

① Purpose:

- ↳ Intendation to achieve.
- ↳ Main Objective.

② Scope:

- ↳ Deliverable./module.
- ↳ Limitation.

③ Boundaries:

- ↳ Limit
- ↳ what can we include or exclude.
- ↳ Assumption

④ Functional Requirements:

- ↳ Features.
- ↳ Input: Obtained & Total Marks.
- ↳ Process: Calculations performed to achieve result.
- ↳ Output: Information produced / Final result / Action.

$P \rightarrow S \rightarrow B \rightarrow F$

$I \rightarrow P \rightarrow O$

Example:

A program to calculate Percentage of assesment mark?

① Purpose: Calculate Percentage

② Scope: Calculating Percentages

③ Boundaries: Percentages.
Obtained, Total marks.

④ Functional Requirement:

- ↳ Calculate percentage.
- ↳ Inputs: Obtained / Total Mark
- ↳ Process: $\frac{Ob}{T.M} \times 100\%$
- ↳ Output: Answer is %age.

Example:

A small cafe wants a program that records the names of the customers, the number of meals ordered and the price of each meal.

The program should calculate each customer's total bill and identify the customer with the highest bill invoice. The information will be stored in a text file. The client only wants the customer's name and highest bill displayed.

① Purpose:

Calculate each bill & identify highest bill customer.

② Scope:

Calculating bills, identify highest bill customer.

③ Boundaries:

- ↳ Customer data stored but only want highest bill and customer's name.
- ↳ No need for database / Store only in text file

④ Functional-Requirements.

- ↳ Input: Names, meals, Price
- ↳ Process: calculate bills, compare bill, identify highest bills.
- ↳ Output: highest bill with customer bill.